LAB GUIDELINES

Lab 17 — Asset Management, Documentation, and Licensing

Maps to CompTIA Server+ objectives **2.7 — Asset management and documentation** (labelling, warranty, leased vs. owned, life-cycle, inventory, serial number, asset tag, service manuals, architecture/infrastructure/workflow diagrams, recovery processes, baselines, change management, server configurations, BIA, MTBF, MTTR, RPO, RTO, SLA, uptime requirements, secure documentation storage) **and 2.8 — Licensing concepts** (per-instance, per-user, per-server, per-socket, per-core, site-based, physical vs. virtual, node-locked, signatures, open source, subscription, license vs. maintenance and support, volume licensing, true-up, version compatibility).

Run all commands on https://killercoda.com/playgrounds/scenario/ubuntu

Required software: built-in tools + an offline spreadsheet or CSV editor.

Step 1 — Auto-discover the asset details

```
{
  echo "asset_tag,$(hostname -f)"
  echo "make,$(dmidecode -s system-manufacturer)"
  echo "model,$(dmidecode -s system-product-name)"
  echo "serial,$(dmidecode -s system-serial-number)"
  echo "uuid,$(dmidecode -s system-uuid)"
  echo "bios_version,$(dmidecode -s bios-version)"
  echo "cpu,$(lscpu | awk -F: '/Model name/{print $2}' | xargs)"
  echo "ram_gb,$(free -g | awk '/Mem/{print $2}')"
  echo "os,$(. /etc/os-release; echo $PRETTY_NAME)"
  echo "ip,$(hostname -I | awk '{print $1}')"
  echo "discovered_at,$(date -Is)"
} > /root/asset-$(hostname).csv
cat /root/asset-$(hostname).csv
```

This is the **inventory** + **make/model** + **serial number** + **asset tag** information from objective 2.7.

Step 2 — Life-cycle stages

Stage	Activity
Procurement	RFQ, vendor selection, HCL check, PO, warranty terms
Deployment	Rack, image, configure, baseline
Usage	Patch, monitor, capacity-trend, change-manage
End of life	De-commission, secure wipe (Lab 25)
Disposal/recycle	Certified e-waste, audit trail, asset retirement

Track which stage each asset is in inside the spreadsheet.

Step 3 — Documentation set the exam expects

Build (or link to) at least these documents per server:

1. **Service manual** — vendor PDF link.
2. **Architecture diagram** — high-level component layout (use draw.io / Excalidraw).
3. **Infrastructure diagram** — network + storage paths.
4. **Workflow diagram** — request → approval → deploy.
5. **Recovery processes** — runbook for power, OS, app failure.

6. **Baselines** — file from Lab 12.
7. **Change management log** — ticket / RFC numbers.
8. **Server configuration** — the running config exported.

Capture the running config as a starting "as-built" doc:

```
mkdir -p /root/asbuilt && cd /root/asbuilt
cp -p /etc/fstab fstab.txt
ip -j addr show > ip-addr.json
systemctl list-units --type=service --state=running > services.txt
ufw status numbered > firewall.txt 2>/dev/null
crontab -l > crontab.txt 2>/dev/null
ls
```

Step 4 — BIA-driven targets (MTBF, MTTR, RPO, RTO, SLA, uptime)

Metric	Means	Driven by
BIA	Business Impact Analysis	Stakeholders
MTBF	Mean Time Between Failures (reliability)	Hardware specs, history
MTTR	Mean Time To Recover	Runbook quality, spares
RPO	Recovery **Point** Objective — max acceptable data loss	Backup frequency (Lab 26)
RTO	Recovery **Time** Objective — max acceptable downtime	DR plan (Lab 27)
SLA	Service Level Agreement — contractual uptime %	Above + redundancy (Lab 14)
Uptime	"Five nines" = 99.999 % ≈ 5.26 min downtime / year	Cluster + DR + change mgmt

Uptime maths cheat:

Uptime %	Annual downtime
99 %	3 d 15 h 36 m
99.9 %	8 h 45 m 36 s
99.99 %	52 m 33 s
99.999 %	5 m 15 s

Step 5 — Change management entry (template)

```
RFC-2026-0521-001
System:         srv01 (asset tag SRV01-2024)
Change window:  2026-05-21 22:00–23:00 SGT
Risk:           Low
Implementer:    alice
Approver:       Bob (CAB)
Rollback:       LVM snapshot lv_root_pre_patch
Description:    Apply unattended-upgrades 22.04.x security set
```

Keep these alongside the asset record.

Step 6 — Secure storage of sensitive documentation

Architecture diagrams and recovery runbooks should never be world-readable. Apply Server+ 2.7 guidance:

```
chmod 750 /root/asbuilt
chown root:root /root/asbuilt -R
```

Off-site copy belongs in an encrypted, access-controlled store (the same place as backup encryption keys — Lab 26/27).

Step 7 — Licensing concepts cheat sheet

Model	When it's billed
Per-instance	Per running OS / app instance
Per-concurrent user	Max simultaneous users
Per-server	Per physical or virtual server
Per-socket	Per populated CPU socket
Per-core	Per physical core (Microsoft, Oracle)
Site-based	Whole site, unlimited within it
Subscription	Recurring fee, includes updates
Volume licensing	Discounted bulk pre-purchase
Open source	No licence cost; obligations per OSS licence (GPL, BSD, MIT…)
Node-locked	Bound to a specific machine ID

Other terms to recognise:

- **Signatures** — cryptographic proof of a licence file's authenticity.
- **License vs. maintenance & support** — separate line items; you can keep using software whose support has lapsed but no patches.
- **True-up** — annual reconciliation of "what you actually used" vs. "what you bought".
- **Version compatibility** — backward compatible (new licence honours old version) vs. forward compatible (old licence honours new version).
- **Physical vs. virtual** — many enterprise licences require declaring whether a host is bare-metal or a VM (often costs differ).

Step 8 — License count validation script

```
cat > /root/license-count.sh <<'EOF'
#!/usr/bin/env bash
echo "Physical sockets: $(lscpu | awk '/Socket\(s\)/{print $2}')"
echo "Physical cores:   $(lscpu | awk '/^Core\(s\) per socket/{c=$NF} /Socket\(s\)/{s=$NF} END{prin
echo "vCPUs:            $(nproc)"
echo "OS instance count on this host: 1"
EOF
chmod +x /root/license-count.sh && /root/license-count.sh
```

Use the output during a **true-up** reconciliation.

Free online tools

- **GLPI (open-source ITAM)** — https://glpi-project.org/
- **Snipe-IT (open-source asset tracking)** — https://snipeitapp.com/
- **draw.io / diagrams.net** — https://app.diagrams.net/
- **Excalidraw** — https://excalidraw.com/
- **CompTIA BCP/DR templates** — https://www.ready.gov/business/implementation

What you learned

- A scripted asset-discovery pattern that fills the inventory spreadsheet.
- The eight documentation artefacts Server+ 2.7 requires.
- BIA-driven metrics: MTBF, MTTR, RPO, RTO, SLA, and the uptime maths.
- A change-management entry template you can carry to interview / exam questions.
- Why sensitive docs need restricted storage.
- The full Server+ 2.8 licensing model table and a quick true-up validation script.